Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 16

Equivalence Relations

Covers Chartrand & Zhang, Section 5.2

Status of this set

Ungraded practice. Solutions released with it. Write the proofs in *complete sentences* — that is the skill being practised.

Part I: How to Practice This Material

Three proofs, clearly labelled

The template

Reflexive. Let $a \in A$. . . . Therefore $a \; R \; a$.
Symmetric. Assume $a \; R \; b$. . . . Therefore $b \; R \; a$.
Transitive. Assume $a \; R \; b$ and $b \; R \; c$. . . . Therefore $a \; R \; c$.
Conclude. Since R is reflexive, symmetric, and transitive, it is an equivalence relation.

The recurring move

In the transitive part you are handed two facts about a, b and b, c and asked for a fact about a, c. **Combine the hypotheses — usually by adding — and let b cancel.**
That single move handles most parity and divisibility relations you will meet.

Three mistakes to expect

1. **Proving two properties and stopping.** All three are required.
2. **Assuming what is to be proved.** In the symmetric part you may use $a \; R \; b$; you may not use $b \; R \; a$.
3. **Writing "$2k$" without saying k is an integer.** The definition of even requires it, and the last line of every parity proof depends on it.

Part II: Practice Problems

Section A — Is it an equivalence relation?

1. For each relation on $\mathbb{Z}$, decide whether it is an equivalence relation. Prove it or name the failing property with a witness.

(a) $a \mathrel{R} b$ if $a - b$ is divisible by 5.

(b) $a \mathrel{R} b$ if $a \cdot b$ is even.

(c) $a \mathrel{R} b$ if $a^2 = b^2$.

Section B — Full proofs

2. Prove that the relation R on $\mathbb{Z}$ defined by $a \mathrel{R} b$ if $a + 3b$ is even is an equivalence relation.

3. Let S be a nonempty set and let R_1 and R_2 be equivalence relations on S. Prove that $R_1 \cap R_2$ is an equivalence relation on S.

Section C — Recognising them

4. Which of these are equivalence relations? One sentence each.

(a) "Has the same birthday as," on the set of people.

(b) "Is a sibling of," on the set of people.

(c) "Has the same file extension as," on a set of filenames.

(d) "Is logically equivalent to," on compound statements.

Section D — Writing

5. In five to eight complete sentences, explain why proving a relation is an equivalence relation requires three separate arguments, and why showing it is *not* one requires only one. Connect this to something you proved earlier in the term.

Section E — Challenge (optional)

6. Show that the union of two equivalence relations need not be an equivalence relation. Give a specific example and say exactly which property fails.